

USER MANUAL

RPV WATER DELIVERY SYSTEM

Administrator and Driver User Guide

1. Introduction

The **RPV Water Delivery System** is a web-based platform designed to help the administrator manage water delivery activities and monitor business performance. The system provides an Administrator Dashboard where delivery assignments can be created, drivers can be registered, delivery statuses can be monitored, and completed delivery records can be reviewed through the history section.

The system also presents business information such as total income, fuel expenses, net profit, delivery completion rate, customer records, registered drivers, and monthly performance. The dashboard includes a **Revenue & Profit Analytics** section that allows the administrator to view business performance using different chart formats.

2. User Roles

The system uses different access levels for its users.

Administrator

The Administrator is responsible for managing the water delivery operation.

The Administrator can:

- Access the Administrator Dashboard
- Add/Register drivers
- Assign new deliveries
- View delivery status
- View delivery history
- Search delivery history
- Monitor revenue
- Monitor fuel costs
- Monitor net profit
- Monitor completion rate
- Review business performance
- View revenue and profit analytics
- Log out of the system

Driver

The Driver receives delivery assignments from the Administrator and is responsible for responding to assigned delivery tasks and completing the required delivery activities.

3. Logging In

The first step in using the system is accessing the appropriate user account.

Steps

1. Open a web browser.
2. Navigate to the system.
3. Enter the registered login credentials.
4. Submit the login form.
5. After successful authentication, the system redirects the user to the appropriate dashboard.

The Administrator should use the Administrator account when managing deliveries and business information.

4. Administrator Dashboard

After logging in, the Administrator is presented with the **Admin Dashboard**.

The dashboard is designed to manage deliveries and monitor business performance. The interface contains a sidebar with the following options:

1. **Dashboard**
2. **Add Driver**
3. **Delivery Status**
4. **History**
5. **Logout**

The dashboard also provides quick business insights and delivery-management functions.

5. Dashboard Quick Insights

The Quick Insights section provides the Administrator with an overview of important business information.

The implemented dashboard displays the following:

Revenue

The **Revenue** indicator displays the total income generated from recorded deliveries.

This allows the Administrator to quickly determine the amount of income generated by the business.

Fuel Cost

The **Fuel Cost** indicator displays the recorded fuel expenses associated with transportation.

This allows the Administrator to monitor the amount spent on fuel.

Net Profit

The **Net Profit** indicator displays earnings after fuel costs.

This provides the Administrator with a quick view of the business's remaining earnings after the recorded fuel expenses.

Completion Rate

The **Completion Rate** shows the percentage of deliveries that have been completed.

The system calculates this using the number of completed deliveries compared with the total number of deliveries.

6. Business Overview

The **Business Overview** section provides additional information about the company's delivery operations.

The implemented system displays:

- **Total Deliveries**
- **Registered Drivers**
- **Customers**
- **Peak Month**
- **Lowest Month**
- **This Month**

The **Peak Month** identifies the month with the highest profit, while **Lowest Month** identifies the month with the lowest profit. The **This Month** value provides the current month's financial information available in the system.

7. Assign New Delivery

The **Assign New Delivery** section is used by the Administrator to create a delivery assignment.

The actual system form includes fields for:

- Customer Name
- Delivery Address
- Driver selection

The system also associates the selected Driver with the assignment.

Steps to Assign a Delivery

1. Go to the **Administrator Dashboard**.
2. Locate **Assign New Delivery**.
3. Enter the **Customer Name**.
4. Enter the customer's **Delivery Address**.
5. Select a Driver from the available driver list.
6. Review the information.
7. Submit the delivery assignment.

When a Driver is selected, the system automatically obtains the corresponding driver name for the assignment.

Important Reminder

Before submitting an assignment, verify:

- Customer name
- Complete delivery address
- Correct Driver

Incorrect information may affect delivery records and business reports.

8. Add Driver

The **Add Driver** option allows the Administrator to register a Driver in the system.

The implemented form requires:

- Full Name
- Address
- Contact Number

The form provides a **Register Driver** button for submitting the Driver information.

Steps to Register a Driver

1. Select **Add Driver** from the Administrator sidebar.
2. The Driver registration window will appear.
3. Enter the Driver's **Full Name**.
4. Enter the Driver's **Address**.
5. Enter the Driver's **Contact Number**.
6. Review the information.
7. Select **Register Driver**.

The newly registered Driver can then become available for delivery assignment.

Closing the Add Driver Window

If the Administrator opens the Add Driver window but does not want to continue:

1. Select the **X/Close** option.
2. The window will close.
3. The Administrator returns to the dashboard.

The system's code specifically implements an Add Driver modal that can be opened and closed.

9. Delivery Status

The **Delivery Status** function allows the Administrator to monitor the status of delivery assignments.

The implemented status table displays:

- Customer
- Driver
- Delivery Date

- Status

The system recognizes delivery statuses including **Accepted** and **Declined**.

Steps

1. Select **Delivery Status** from the sidebar.
2. The Delivery Status window will appear.
3. Review the list of deliveries.
4. Check the assigned Customer.
5. Check the Driver.
6. Check the Delivery Date.
7. Review the current Status.

This allows the Administrator to determine whether assigned deliveries have been accepted or declined.

Closing Delivery Status

Select the close button on the window to return to the Administrator Dashboard.

10. Delivery History

The **History** function allows the Administrator to review previously recorded delivery information.

The actual system history table displays:

- Customer
- Location
- Driver
- Assigned Date
- Delivery Date
- Income
- Profit

The income is displayed using the recorded **total price**, while the profit is displayed using the recorded **profit** value.

Steps

1. Select **History** from the Administrator sidebar.
 2. The Delivery History window will appear.
 3. Review the available delivery records.
 4. Check the customer.
 5. Check the delivery location.
 6. Check the assigned Driver.
 7. Review the assigned and delivery dates.
 8. Review the income.
 9. Review the profit.
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11. Searching Delivery History

The Delivery History section includes a search function.

This allows the Administrator to locate a specific record without manually reviewing every entry.

Steps

1. Open **History**.
2. Locate the **Search** field.
3. Enter a keyword.
4. The system searches the displayed delivery rows.
5. Records that match the entered text remain visible.

The search function checks the text contained in the delivery-history rows, allowing the Administrator to search information such as customer, location, Driver, dates, income, or profit.

Example

If the Administrator enters a customer's name in the search field, the system displays the history records containing that name.

12. Revenue and Profit Analytics

The Administrator Dashboard includes a **Revenue & Profit Analytics** section.

This section presents business performance over time using a chart.

The implemented system allows the Administrator to select among:

- **Line**
- **Bar**
- **Pie**

chart formats.

Steps

1. Go to the Administrator Dashboard.
2. Scroll to **Revenue & Profit Analytics**.
3. Locate the chart-type selection.
4. Select **Line**, **Bar**, or **Pie**.
5. Review the displayed business performance information.

The chart uses the system's accumulated profit information to provide a visual representation of business performance over time.

13. Monitoring Business Performance

The Administrator can use the dashboard to monitor several business indicators without manually calculating them.

The dashboard provides information about:

- Total income
- Fuel cost
- Net profit
- Completion rate
- Total deliveries
- Registered drivers
- Customers
- Peak month

- Lowest month
- Current-month performance

These indicators provide the Administrator with a quick overview of the business's delivery and financial performance.

14. Understanding the Financial Information

Revenue

Revenue represents the total income generated from recorded deliveries.

Fuel Cost

Fuel Cost represents the transportation expense recorded for fuel.

Net Profit

Net Profit represents the earnings after the recorded fuel costs.

Completion Rate

Completion Rate represents the percentage of completed deliveries compared with the total number of deliveries.

Peak Month

Peak Month identifies the month with the highest recorded profit.

Lowest Month

Lowest Month identifies the month with the lowest recorded profit.

These indicators help the Administrator understand the financial and operational condition of the business.

15. Monitoring Delivery Information

The Administrator can monitor delivery records using both **Delivery Status** and **History**.

Delivery Status

Used to check the current status of delivery assignments.

History

Used to review previously recorded delivery information and financial results.

The combination of these functions allows the Administrator to monitor both current delivery activity and historical performance.

16. Firebase Data Storage

The system uses **Firebase Firestore** for storing and retrieving the delivery-related data.

The code shows Firebase Firestore being initialized and used to retrieve collection data through `onSnapshot`, allowing the dashboard information to reflect stored records.

The delivery records contain information used by the dashboard, including values such as:

- Customer Name
- Location
- Driver
- Assigned Date
- Delivery Date
- Total Price
- Fuel Cost
- Profit
- Status

These records provide the underlying information used for delivery monitoring and business analytics.

17. Logging Out

The **Logout** option allows the Administrator to safely exit the system.

Steps

1. Locate the sidebar.
2. Select **Logout**.
3. The system redirects the user away from the Administrator Dashboard.

The implemented system provides a dedicated logout route for this function.

Users should always log out after completing their work, especially when using a shared computer.

18. Quick Administrator Workflow

The basic Administrator workflow is:

Login

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Administrator Dashboard

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Add Driver *(if a new Driver needs to be registered)*

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Assign New Delivery

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Monitor Delivery Status

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Review Delivery History

↓

Monitor Revenue, Fuel Cost, and Net Profit

↓

Review Completion Rate and Business Overview

↓

Analyze Revenue & Profit Chart

↓

Logout

19. Important Data Entry Reminders

The Administrator should verify information before submitting a delivery assignment.

Customer Information

Ensure that the customer's name is entered correctly.

Delivery Address

Enter the complete delivery address to provide the Driver with accurate delivery information.

Driver

Select the correct Driver before submitting the delivery.

Financial Information

Review recorded income, fuel cost, and profit information when checking delivery history.

Accurate information is important because these records are used to calculate and display the dashboard's business indicators.

20. Troubleshooting

Problem	Recommended Action
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Cannot log in	Check the account credentials and try again.
Driver is not available for selection	Check whether the Driver has been registered through Add Driver .
Incorrect customer information	Review the delivery record and correct the information using the available system function.
Incorrect delivery address	Verify the address before assigning the delivery.
Delivery status is not as expected	Open Delivery Status and verify the assigned Driver and delivery record.
History record cannot be found	Use the Search field in Delivery History.
Dashboard information appears incorrect	Check the underlying delivery records and verify that the required information has been recorded correctly.
Analytics chart is difficult to interpret	Change the chart type between Line, Bar, and Pie.
System does not load	Check the internet connection and refresh the browser.
Unable to exit the system	Use the Logout option from the sidebar.

21. Mobile and Responsive Use

The system interface includes responsive behavior for smaller screens. When the screen width becomes smaller, the sidebar is hidden and the dashboard content changes to a single-column layout.

This allows the system to be accessed through smaller devices such as smartphones while maintaining access to the main dashboard functions.

Users should use a modern mobile browser and maintain a stable internet connection when accessing the system from a phone.

22. System Limitations Based on the Implemented Code

The user manual should accurately reflect what is **actually implemented** in the system.

Based on the provided code, the Administrator interface currently focuses on:

- Dashboard monitoring
- Driver registration
- Delivery assignment
- Delivery status
- Delivery history
- Search within history
- Revenue monitoring
- Fuel-cost monitoring
- Net-profit monitoring
- Completion-rate monitoring
- Business overview
- Revenue and profit analytics
- Logout

Therefore, functions such as **GPS tracking, real-time vehicle tracking, detailed odometer recording, separate fleet management, route optimization, and cost prediction** should not be presented in the user manual as currently available features unless they are implemented elsewhere in your actual project code.

That distinction is important for your capstone because the **user manual should describe the system the client can actually use**, not only the functions described in the proposed workflow.

23. Conclusion

The RPV Water Delivery System provides the Administrator with a centralized interface for managing water delivery assignments and monitoring business performance. The Administrator can register Drivers, assign deliveries, monitor delivery statuses, review delivery history, and examine important financial indicators such as revenue, fuel cost, and net profit.

The dashboard also provides visual analytics that allow the Administrator to examine business performance through different chart formats. By using the system consistently and entering accurate delivery information, the client can maintain organized delivery records and obtain a clearer view of business performance.

For the final capstone user manual, screenshots of the actual login page, Administrator Dashboard, Add Driver window, Assign New Delivery form, Delivery Status window, History window, and Analytics section should be inserted in the corresponding sections.